CS:330 Final Project Reflection

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There were a few notable development choices that I made for this 3D scene. I received feedback for my original project proposal, suggesting that I reduce the scope of the objects to be replicated to that of the microwave and the items on top. This was a good choice and allowed me to focus on the accuracy of the scale, position, and rotation of the objects. This also gave me time to give the scene good lighting and the correct textures and materials for each object. Up until the final week, I used the microwave, mug, and scene backdrops to practice with proper textures, lighting, and materials before working on the rest of the objects. The backdrops and counter were simple to position. I decided to use different textures for the walls and counters to make the scene more interesting. The tile texture on the wall needed a material that was shinier than wall paint, but duller than normal ceramic, so I changed the diffuse, specular, and shininess in the definition for the material. For the mug objects, they consisted of a cylinder with no top, a torus for the lip, and a rotated half torus for the handle. Further in development I added a second inner-cylinder with no top or bottom to give the inside a different texture from the outside. The mushroom cup was based off this, but with a thick half-sphere placed on top.

The virtual camera for the 3D scene allows the user to use the WASD keys, the Q&E keys, the O&P, and the scroll wheel to navigate the scene and change to different views. A&D pans the camera left and right. W&S keys bring the camera forward and backward in the scene. The Q&E keys allow for vertical movement along the y-axis. Each of these camera functions are influenced by a speed variable, which can be altered by the scroll wheel within a fixed range, allowing for quick or slow movements about the scene, Finally, the O key switches the view to an orthographic view for a true-to-scale scene, and the P key switches the view to the default perspective scene.

One thing I did to make the code more organized, and modular was rendering each object from its own specific function, rather than rendering each mesh in the same function. The RenderScene() function simply called a function for each object. For example, it would call the RenderMushroomCup() function and continue to transform and draw the object into the scene. This structure for rendering the objects allows me to easily define a new function for an object or alter an existing one. It also allowed me to easily duplicate an object to use as a base for another, such as how I used the RenderCupOne() function to build the RenderMushroomCup() function. While this was a good start, I think a larger scene with more objects would need a function that simplified changing the transformations. A bit of my time was spent manually altering the transformations of the individual's basic shapes when it would save time to be able to move each whole object together or scale evenly.